

Redes Neuronales Convolucionales (CNN)

Red neuronal multicapa

Convolutional
(Preprocesamiento
del hiperespacio
de entrada)

Aplando de la
información
(flatten)

Red neuronal
multicapa

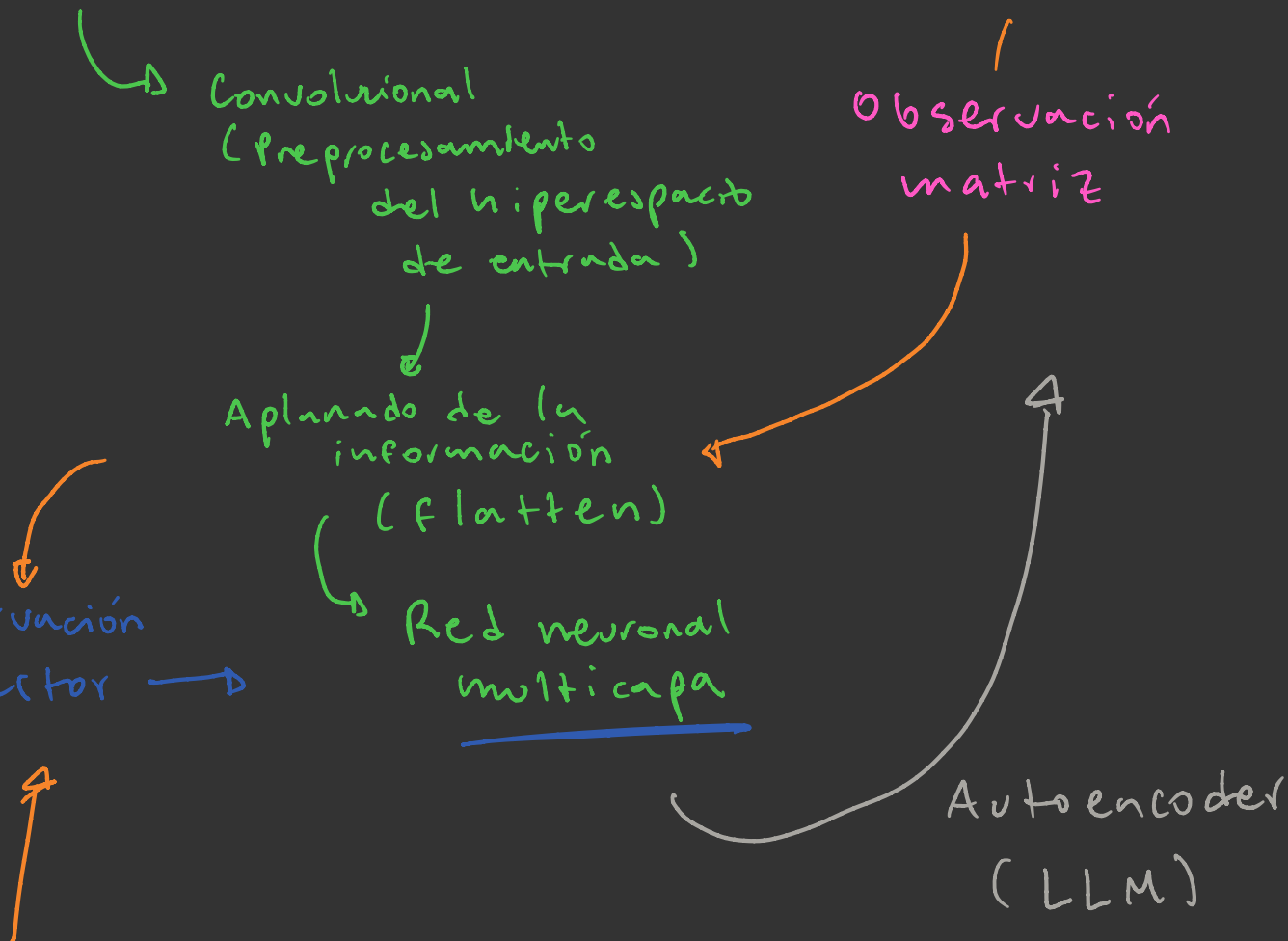
Observación
vector →

imagen

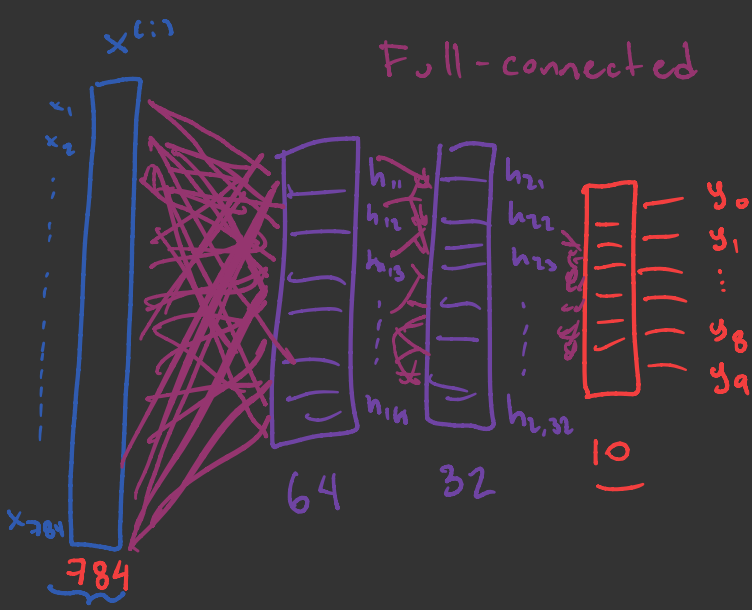
Imagen, Video

Observación
matriz

Autoencoder
(LLM)



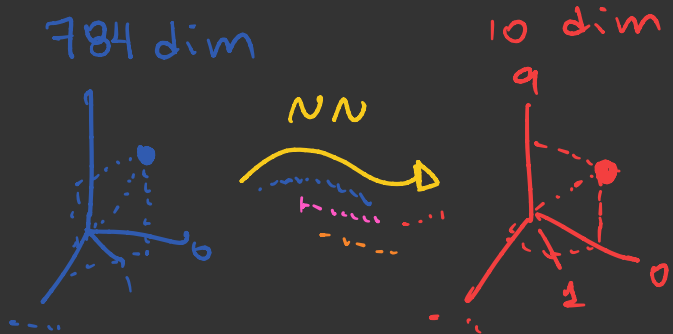
Red neuronal multicapa



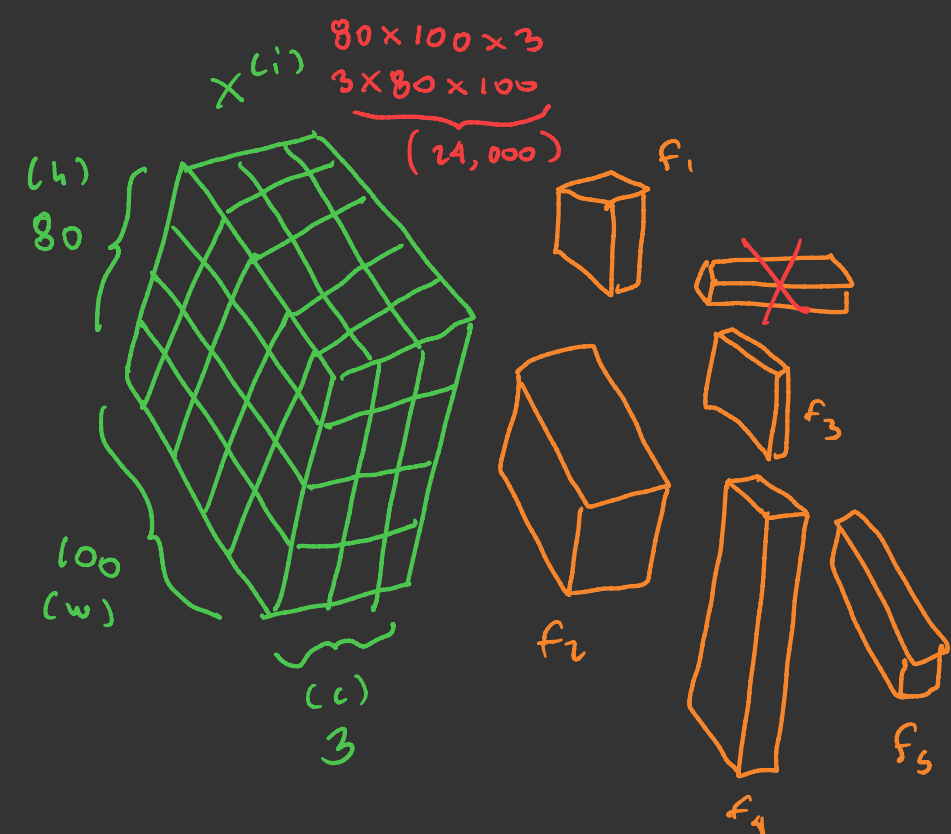
28x28 pixels
información
aplanada

$$X_a = \begin{pmatrix} x^{(1)} & 1 & 2 & \dots & 784 \\ x^{(2)} & & & & \\ \vdots & & & & \\ x^{(10,000)} & & & & \end{pmatrix} \quad (i)$$

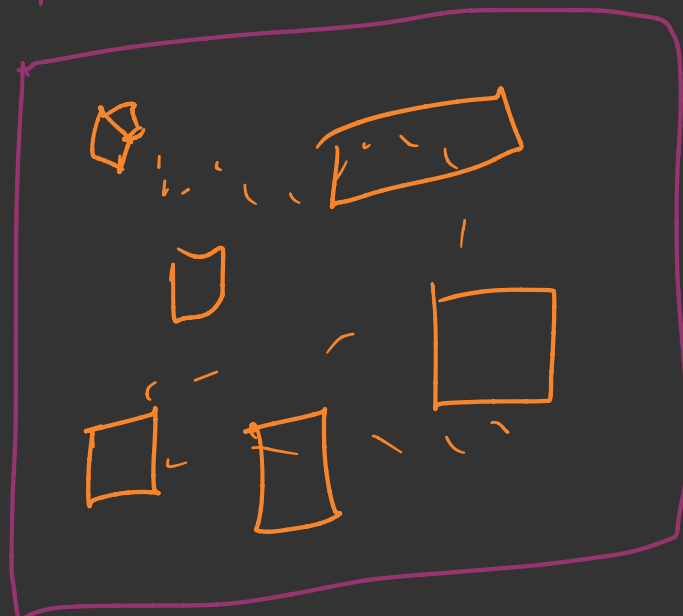
$$y_a = \begin{pmatrix} y^{(1)} & 1 & 2 & \dots & 9 \\ y^{(2)} & & & & \\ \vdots & & & & \\ y^{(10,000)} & & & & \end{pmatrix}$$



Red Neuronal Convocional



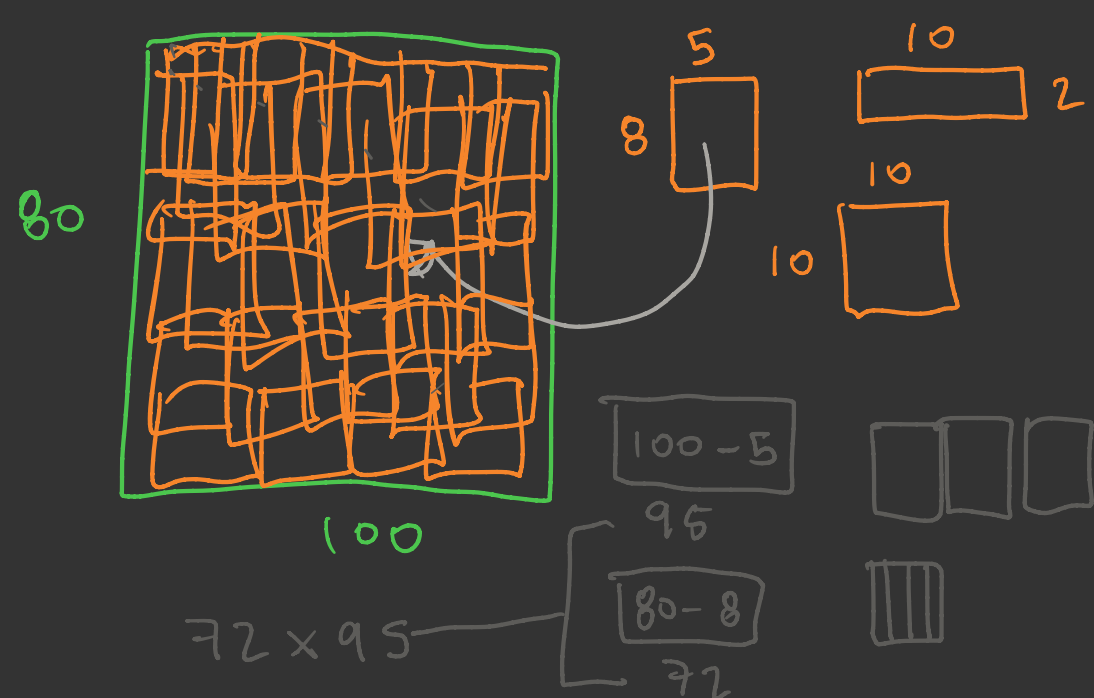
Max Pooling 500



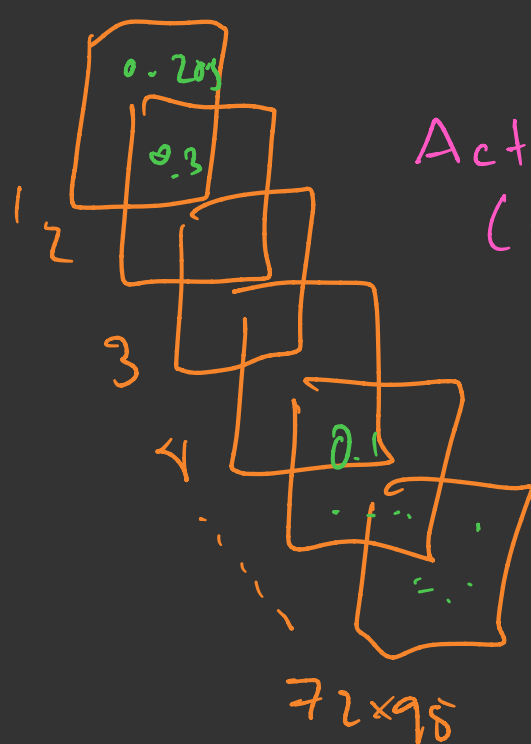
D_0

D_p

Flatten 20



$f_1(p)$



Activación (ReLU)

+ Filtro ponderado (β)

$ReLU(x \cdot \beta) \geq 0$

Operador convolucional

- Input (...)
 - Conv2D(...)
 - Max Pooling(...)
- Entrada {
 - Flatten(...)
- Ocultas {
 - Dense(...)
 - Dense(...)
- Salida {
 - Dense(...)